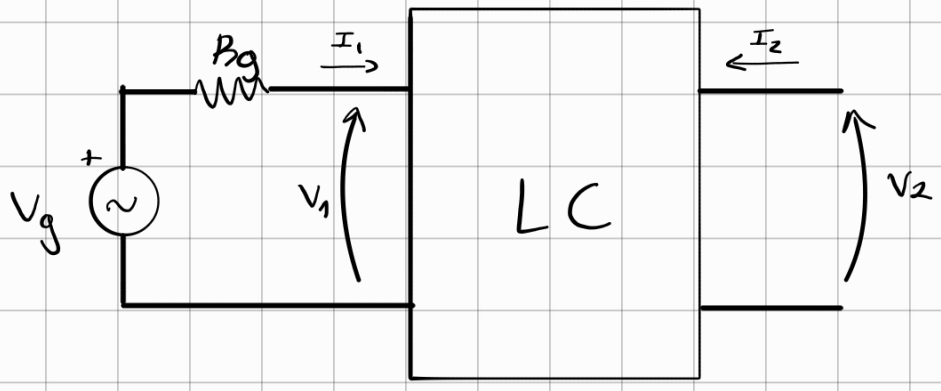
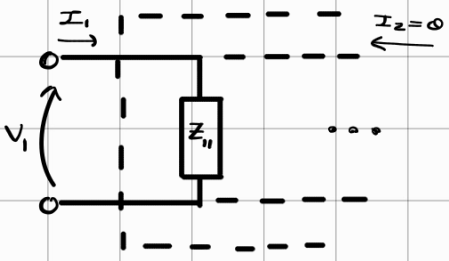
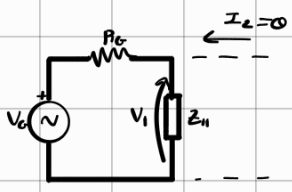
1)



- $T(s) = \left.\frac{V_2}{V_G}\right|_{I_2=0} = \left.\frac{V_1}{V_g}\right|_{I_2=0} \cdot \left.\frac{V_2}{V_1}\right|_{I_2=0}$

↳ Se propone un cuadripolo:





$\Longrightarrow \quad \left.\frac{V_1}{V_g}\right|_{I_2=0} = \frac{Z_{11}}{Z_{11}+R_G}$ //

↳ $\left.\frac{V_2}{V_1}\right|_{I_2=0} = \frac{1}{A} = \frac{Z_{21}}{Z_{11}}$ //

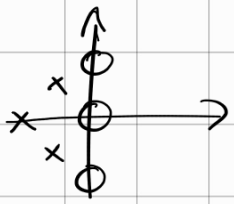
$\Longrightarrow T(s) = \left.\frac{V_1}{V_g}\right|_{I_2=0} \cdot \left.\frac{V_2}{V_1}\right|_{I_2=0} = \frac{Z_{11}}{Z_{11}+R_G}\,\frac{Z_{21}}{Z_{11}} = \frac{Z_{21}}{Z_{11}+R_G}$

$\Longrightarrow$ $$\boxed{T(s) = \frac{Z_{21}/R_G}{1+Z_{11}/R_G}}$$ //

$$T(s) = \frac{V_2}{V_g} = \frac{K\,s\,(s^2+1/16)}{s^3+2s^2+2s+1} = \frac{K\cdot P}{MQ+NQ}$$

- Se requiere que el Num: $\frac{P(s)}{X(s)}$ sea impar.
  'P(s) = $s(s^2+1/16)$' es impar, entonces $X(s) = M_Q$
  que es par / $M_Q = 2s^2+1$ ∧ $N_Q = s^3+2s$.

$$T(s) = \frac{\frac{P}{MQ}}{1 + \frac{NQ}{MQ}} = \frac{K\,\frac{s(s^2+1/16)}{2s^2+1}}{1 + \frac{s^3+2s}{2s^2+1}}$$

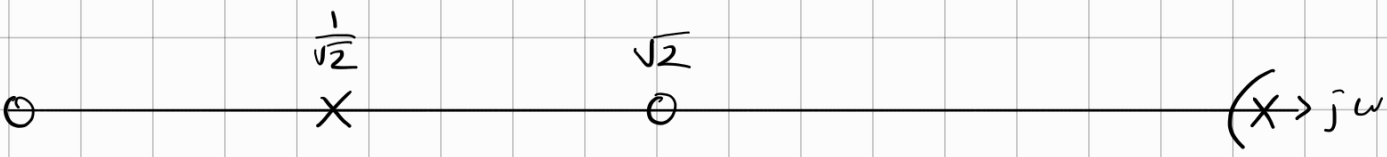


- $\frac{Z_{11}}{R_G} = \frac{s(s^2+2)}{2s^2+1} = \frac{1}{2}\,\frac{s(s^2+2)}{s^2+\frac{1}{2}}$
- $\frac{Z_{21}}{R_G} = \frac{s(s^2+1/16)}{2s^2+1} = \frac{1}{2}\,\frac{s(s^2+1/16)}{s^2+\frac{1}{2}}$

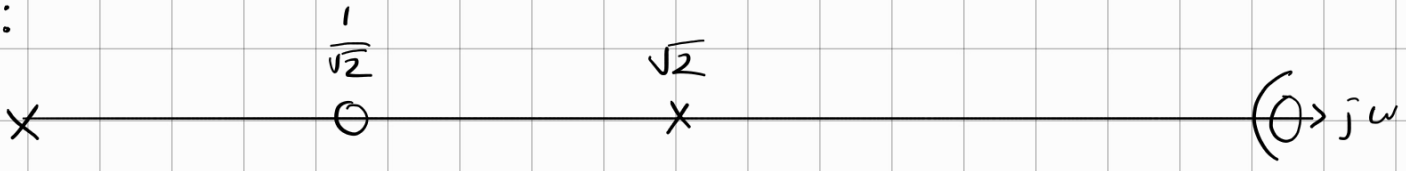
T:



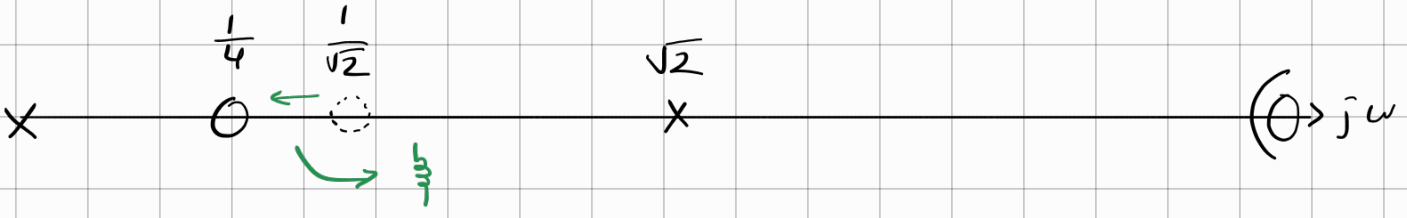
Z_{11}:



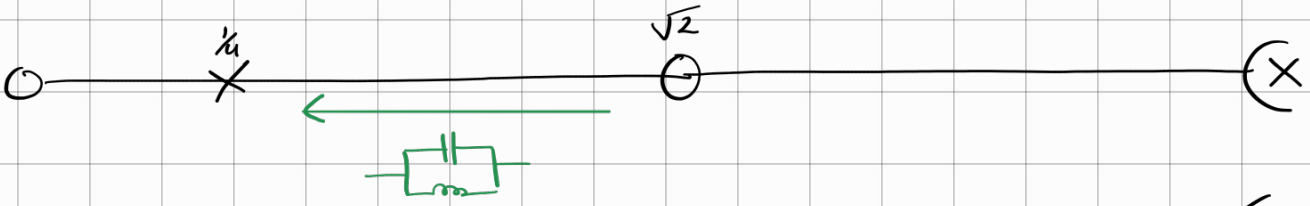
$\frac{1}{Z_{11}}$:



Y_2:



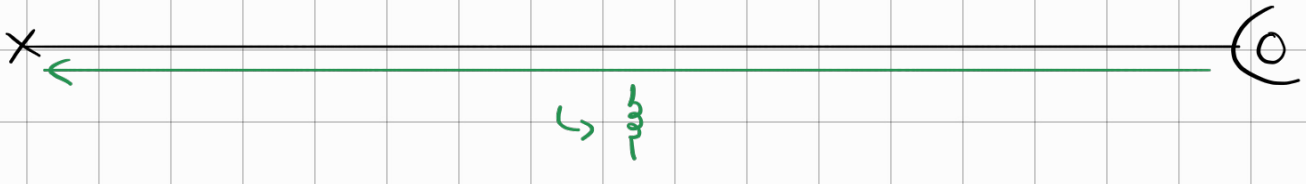
Z_2:



Z_4:



Y_4:



- $\frac{1}{Z_{11}} = \frac{2s^2+1}{s(s^2+2)}$ \

  ↳ $\lim_{s^2 \to -1/16} s \cdot \frac{1}{Z_{11}} = K_0' \rightarrow \lim_{s^2 \to -1/16} \frac{2s^2+1}{s^2+2} = K_0' \Longrightarrow \boxed{K_0' = \frac{14}{31}}$

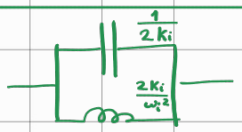
  $\frac{31}{14}$

  ↳ $Y_2 = \frac{1}{Z_{11}} - \frac{14}{31}\frac{1}{s} = \frac{2s^2+1}{s(s^2+2)} - \frac{\frac{14}{31}(s^2+2)}{s(s^2+2)} = \frac{\frac{48}{31}s^2 + \frac{3}{31}}{s(s^2+2)}$

  ↳ $Z_2 = \frac{31}{48} \cdot \frac{s(s^2+2)}{s^2+\frac{1}{16}}$ //

- $2K_i = \lim_{s^2 \to -1/16} \frac{s^2+1/16}{s} \cdot \frac{31}{48} \cdot \frac{s(s^2+2)}{s^2+\frac{1}{16}} \Longrightarrow \boxed{2K_i = \frac{31^2}{3 \times 16^2}}$

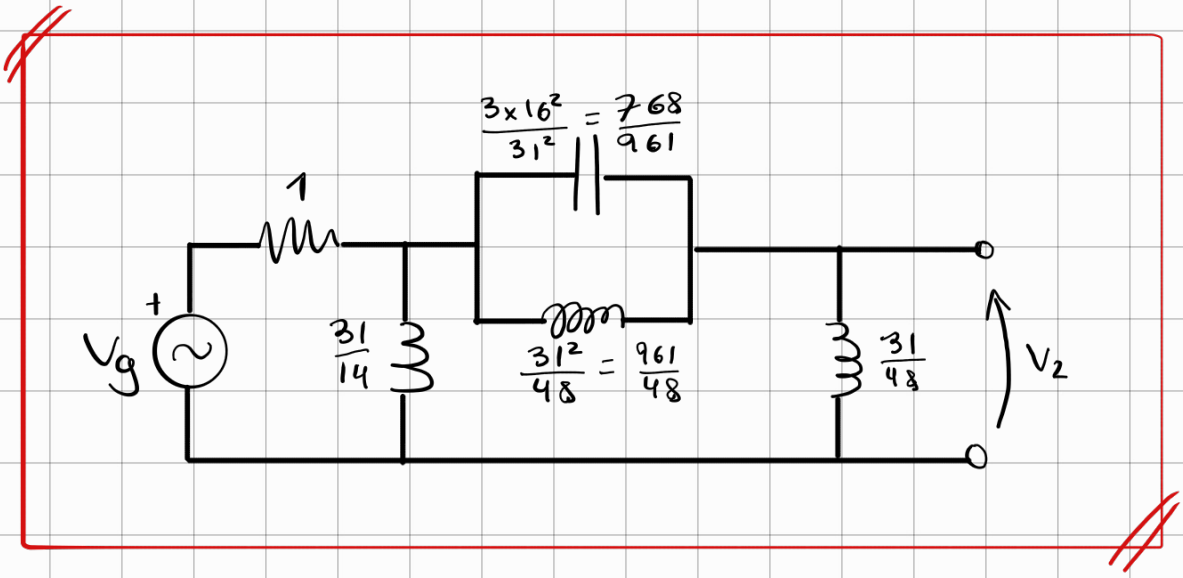
  ↳ $Z_4 = Z_2 - \frac{2K_i s}{s^2+1/16}$

  $= \frac{31}{48} \cdot \frac{s(s^2+2)}{s^2+\frac{1}{16}} - \frac{31}{48} \cdot \frac{31}{16} \cdot \frac{s}{s^2+1/16}$

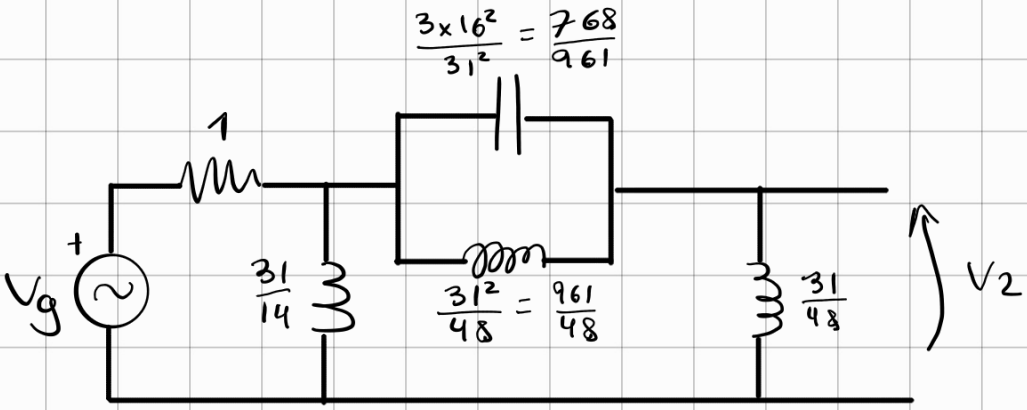
  $= \frac{31}{48} \cdot s \Longrightarrow \boxed{Y_4 = \frac{48}{31}\frac{1}{s}}$

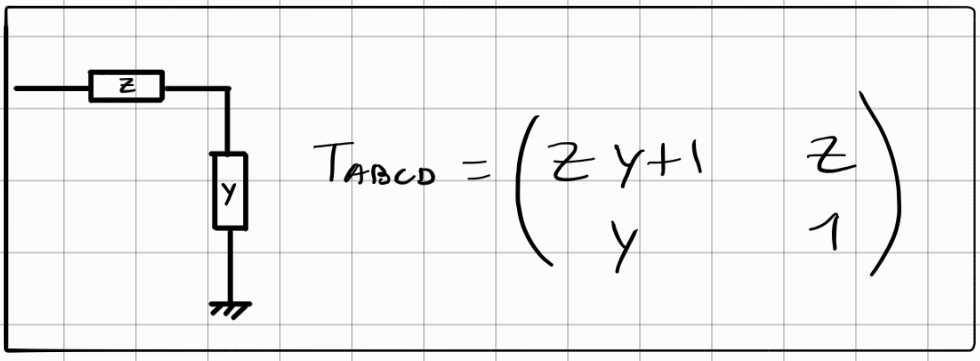
g)



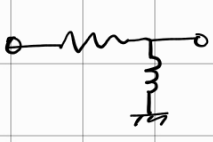
Bonus

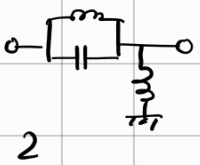
Verificación:





$$T_{ABCD} = \begin{pmatrix} Zy+1 & Z \\ y & 1 \end{pmatrix}$$





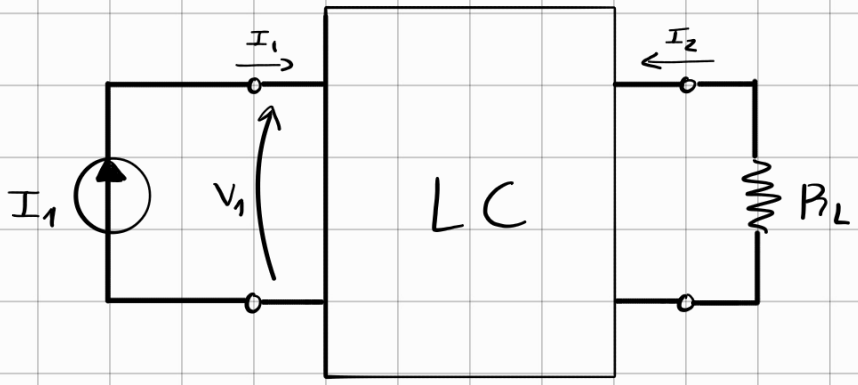
$$T_{TOT} = \begin{pmatrix} \frac{s+\frac{14}{31}}{s} & 1 \\ \frac{14}{31}\,\frac{1}{s} & 1 \end{pmatrix} \cdot \begin{pmatrix} \frac{s^2+2}{s^2+1/16} & - \\ \frac{48}{31}\,\frac{1}{s} & - \end{pmatrix}$$

- $$A = \frac{\left(s+\frac{14}{31}\right)\left(s^2+2\right)}{s\left(s^2+1/16\right)} + \frac{48}{31}\,\frac{s^2+\frac{1}{16}}{s\left(s^2+1/16\right)} = \frac{s^3+2s^2+2s+1}{s\left(s^2+1/16\right)}$$

b)

$$\Rightarrow \quad T(s) = \frac{1}{A} = \frac{K s\left(s^2+1/16\right)}{s^3+2s^2+2s+1} \Big/ K=1$$

2)



$$T(s) = \frac{V_2}{I_1} = \frac{K(s^2+9)}{s^3+2s^2+2s+1}$$

- Utilizamos parámetros Z ya que poseemos un generador de corriente ideal en la entrada y necesitamos que el primer componente esté en derivación.

$$\begin{cases} (-I_2)\,R_L = V_2 \\ V_2 = Z_{21} I_1 + Z_{22} I_2 \end{cases} \Longrightarrow -I_2 = \frac{1}{R_L} V_2$$

$$\Longrightarrow V_2\left(1 + \frac{Z_{22}}{R_L}\right) = Z_{21} I_1$$

$$\Longrightarrow \boxed{T(s) = \frac{V_2}{I_1} = \frac{Z_{21}}{1 + \frac{Z_{22}}{R_L}}}$$

- Parámetro transferencia siempre es impar

$$T(s) = \frac{K(s^2+9)}{s^3+2s^2+2s+1} = \frac{K(s^2+9)}{(s^3+2s)+(2s^2+1)}$$

(Par)

$$= \frac{K\,\frac{s^2+9}{s^3+2s}}{1+\frac{2s^2+1}{s^3+2s}} \Longrightarrow \boxed{\begin{aligned} &\bullet\ Z_{21} = K \cdot \frac{s^2+9}{s(s^2+2)} \\ &\bullet\ Z_{22}/R_L = \frac{2s^2+1}{s(s^2+2)} = 2\,\frac{s^2+\frac{1}{2}}{s(s^2+2)} \end{aligned}}$$

T :



Z_{22} :



$\frac{1}{Z_{22}}$:



Y_2 :



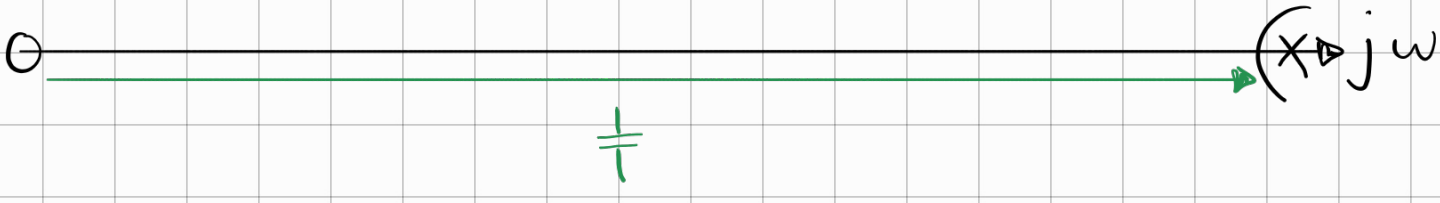
Z_2 :



Z_4 :



Y_4 :



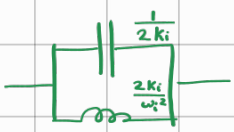
- $Z_{22} = \frac{2R_L s^2 + R_L}{s(s^2+2)} \longrightarrow \frac{1}{Z_{22}} = \frac{s(s^2+2)}{2R_L s^2 + R_L}$ //

- $\lim_{s^2 \to -9} \frac{s(s^2+2)}{2R_L s^2 + R_L} = s K'_\infty \longrightarrow \boxed{K'_\infty = \frac{1}{R_L} \cdot \frac{7}{17}}$ ✓

  ↳ $Y_2 = \frac{s(s^2+2)}{2R_L s^2 + R_L} - \frac{1}{R_L} \cdot \frac{7}{17} s = \frac{3}{17} \frac{s(s^2+9)}{R_L(2s^2+1)}$ //

- $Z_2 = \frac{17}{3} R_L \frac{2s^2+1}{s(s^2+9)}$ //

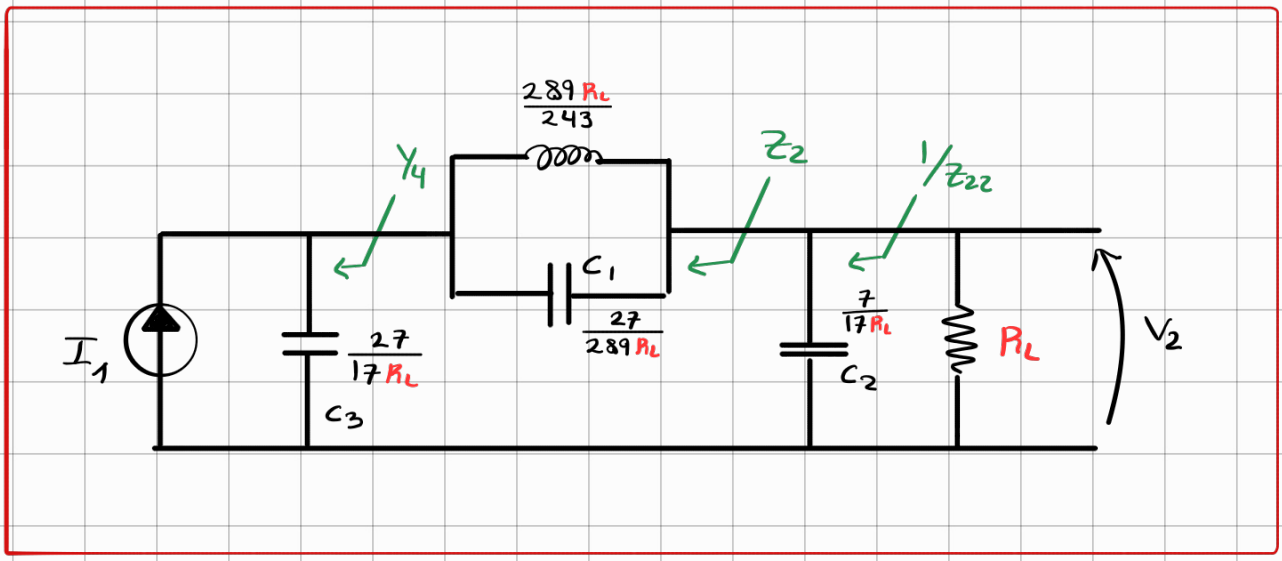
  ↳ $\lim_{s^2 \to -9} \frac{17}{3} R_L \frac{2s^2+1}{s(s^2+9)} = \frac{2k_i s}{s^2+9} \rightarrow \boxed{2k_i = \frac{17^2}{27} R_L}$



  ↳ $Z_4 = \frac{17}{3} R_L \frac{2s^2+1}{s(s^2+9)} - \frac{\frac{17^2}{27} R_L s^2}{(s^2+9)s} = \frac{17}{27} R_L \frac{1}{s}$ //

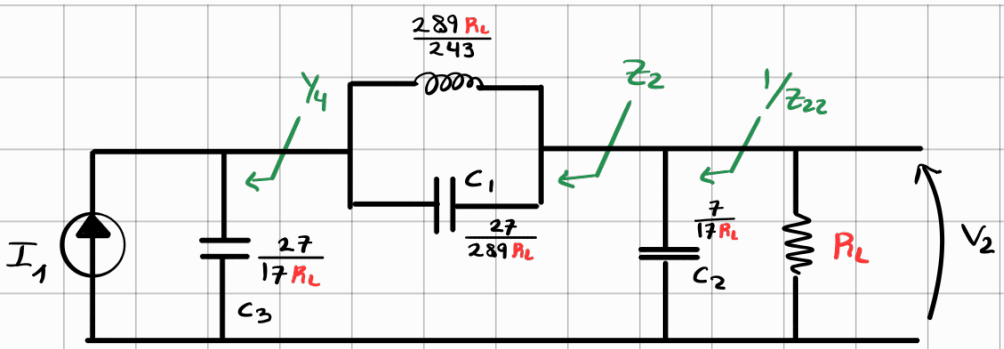
  ↳ $Y_4 = \frac{27}{17} \frac{1}{R_L} s$

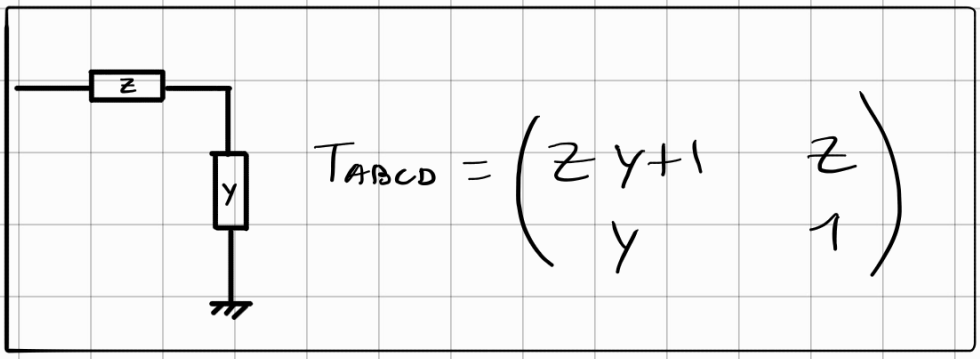
Q)



Bonus

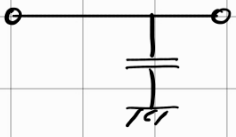
Verificación :

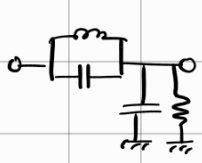




$$T_{ABCD} = \begin{pmatrix} ZY+1 & Z \\ Y & 1 \end{pmatrix}$$

$$\frac{s^2\frac{17\times 7+27}{27} + s\frac{17^2}{27} + 9}{s^2+9}$$





$$\begin{cases} Z_{LC} = \dfrac{s\frac{1}{C_1}}{s^2 + \frac{1}{C_1}} \\ Y_{RC} = sC_2 + \dfrac{1}{R_L} \end{cases}$$

$$T_{TOT} = \begin{bmatrix} - & - \\ \frac{27}{17\,R_L}s & 1 \end{bmatrix} \circ \begin{bmatrix} \frac{\frac{146}{27}s^2 + \frac{289}{27}s + 9}{s^2+9} & - \\ s\frac{7}{17R_L} + \frac{1}{R_L} & - \end{bmatrix}$$

- $$R_L\,C = \left(\frac{27}{17}s\right) \circ \left(\frac{\frac{146}{27}s^2 + \frac{289}{27}s + 9}{s^2+9}\right) + \left(s\frac{7}{17} + 1\right) \circ 1$$

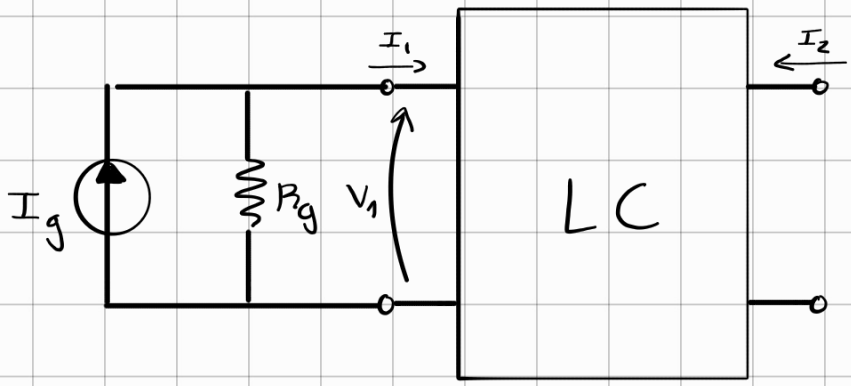
$$= \frac{\frac{146}{17}s^3 + 17s^2 + \frac{243}{17}s + s^3\frac{7}{17} + s^2 + s\frac{63}{17} + 9}{s^2+9}$$

$$= \frac{9s^3 + 18s^2 + 18s + 9}{s^2+9}$$

b)

$$T(s) = \frac{1}{C} = \frac{R_L}{9} \cdot \frac{s^2+9}{s^3 + 2s^2 + 2s + 1} \Big/ K = \frac{R_L}{9}$$

3)



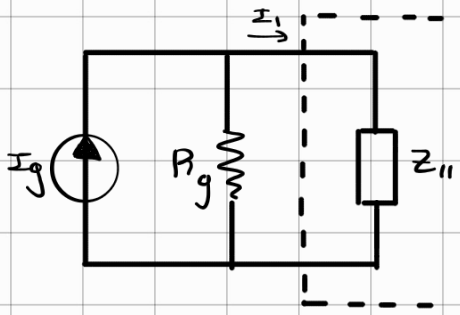
$$T(s) = \frac{V_2}{I_g} = \frac{K \cdot s(s^2 + 1/16)}{s^3 + 2s^2 + 2s + 1}$$

- Utilizamos parámetros Z ya que la condición de medición nos impone $I_2 = 0$ por lo que requerimos que el ultimo componente esté en derivación.

- $$T(s) = \left.\frac{V_2}{I_g}\right|_{I_2=0} = \left.\frac{V_2}{I_1}\right|_{I_2=0} \cdot \left.\frac{I_1}{I_g}\right|_{I_2=0}$$

$$= z_{21} \cdot \frac{1}{1 + \frac{z_{11}}{R_G}}$$

$$\boxed{T(s) = \frac{z_{21}}{1 + z_{11}/R_G}}$$



- $$I_1 = I_g \cdot \frac{1}{1 + \frac{z_{11}}{R_g}}$$

- Parámetro transferencia siempre es impar

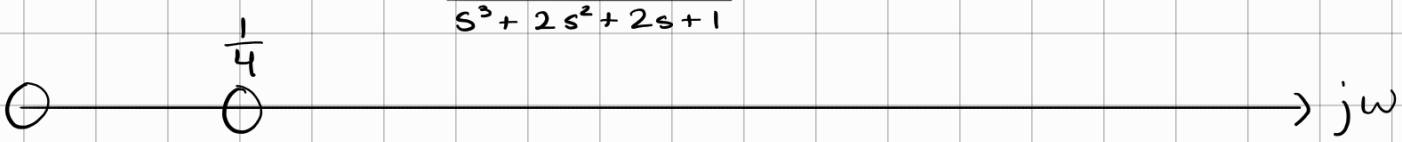
$$T(s) = \frac{K \cdot s(s^2 + 1/16)}{s^3 + 2s^2 + 2s + 1} = K \frac{s^3 + \frac{1}{16}s}{(s^3 + 2s) + (2s^2 + 1)}$$ (impar)

$$= \frac{K \frac{s^3 + \frac{1}{16}s}{2s^2 + 1}}{1 + \frac{s^3 + 2s}{2s^2 + 1}} \Longrightarrow$$

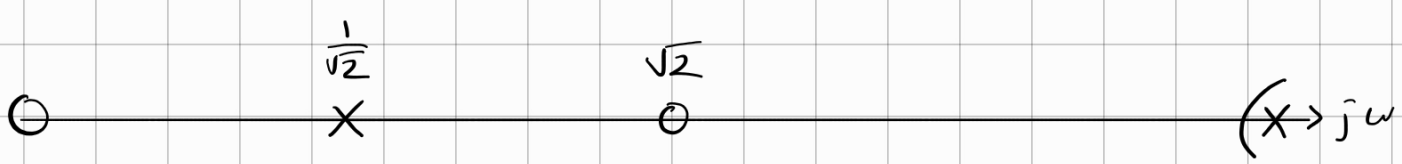
$$\boxed{\begin{aligned} z_{21} &= \frac{s^3 + \frac{1}{16}s}{2s^2 + 1} \\ z_{11} &= R_g \frac{s^3 + 2s}{2s^2 + 1} \end{aligned}}$$

$$\frac{K \cdot s(s^2 + 1/16)}{s^3 + 2s^2 + 2s + 1}$$

T:



Z_{11}:



$\frac{1}{Z_{11}}$:



Y_2:



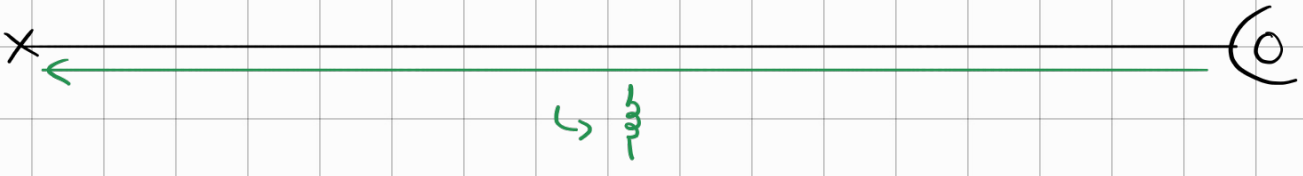
Z_2:



Z_4:



Y_4:



- $\frac{1}{Z_{11}} = \frac{1}{R_g} \frac{2s^2+1}{s(s^2+2)}$

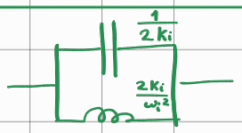
  ↳ $\lim_{s^2 \to -1/16} s \cdot \frac{1}{Z_{11}} = K_0' \rightarrow \lim_{s^2 \to -1/16} \frac{1}{R_g} \frac{2s^2+1}{s^2+2} = K_0' \Rightarrow \boxed{K_0' = \frac{14}{31 R_g}}$

  $\frac{31}{14} R_g$

  ↳ $Y_2 = \frac{1}{Z_{11}} - \frac{14}{31 R_g} \cdot \frac{1}{s} = \frac{2s^2+1}{R_g s(s^2+2)} - \frac{\frac{14}{31}(s^2+2)}{R_g s(s^2+2)} = \frac{\frac{48}{31}s^2 + \frac{3}{31}}{R_g s(s^2+2)}$

  ↳ $Z_2 = \frac{31}{48} R_g \cdot \frac{s(s^2+2)}{s^2+1/16}$ //

- $2K_i = \lim_{s^2 \to -1/16} \frac{s^2+1/16}{s} \frac{31}{48} R_g \frac{s(s^2+2)}{s^2+\frac{1}{16}} \Rightarrow \boxed{2K_i = \frac{31^2}{3 \times 16^2} R_g}$

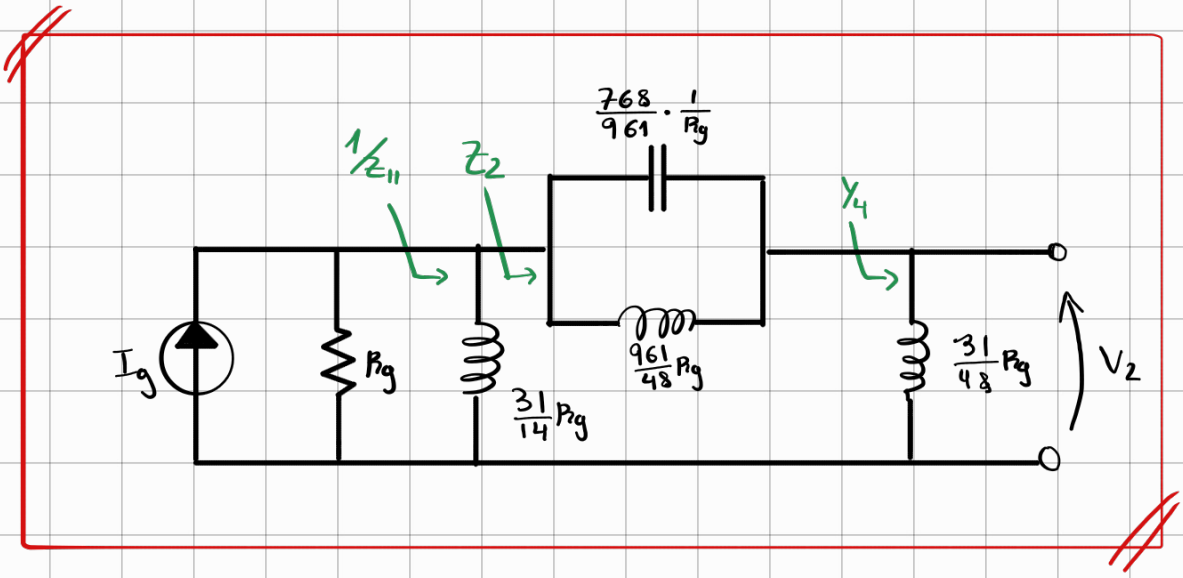
  

  ↳ $Z_4 = Z_2 - \frac{2K_i s}{s^2+1/16}$

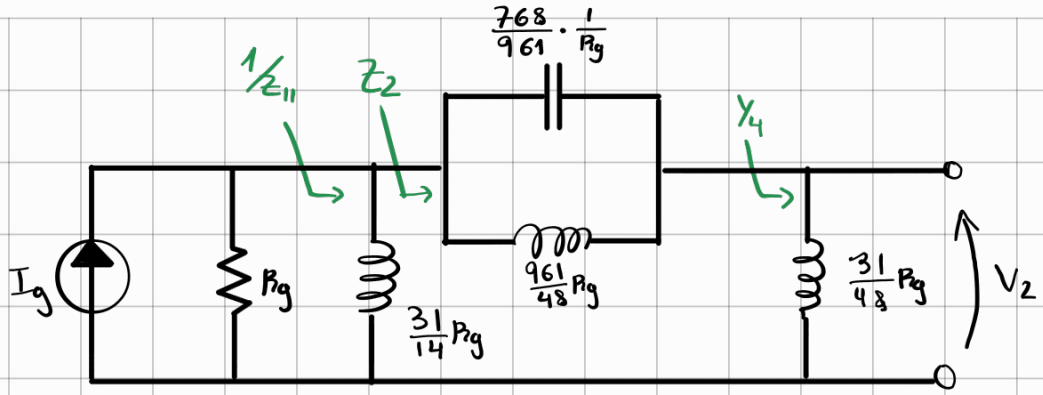
  $= \left(\frac{31}{48} \cdot \frac{s(s^2+2)}{s^2+\frac{1}{16}} - \frac{31}{48} \cdot \frac{31}{16} \frac{s}{s^2+1/16} \right) R_g$

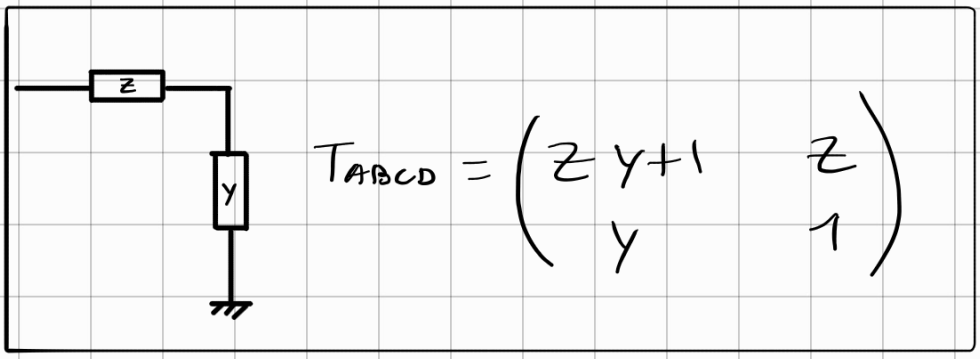
  $= \frac{31}{48} R_g s \Longrightarrow \boxed{Y_4 = \frac{48}{31} \frac{1}{R_g} \frac{1}{s}}$

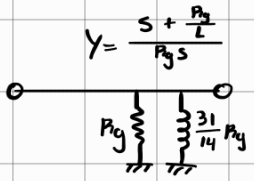
9)

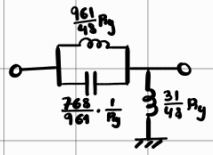


Verificación:









$$\begin{cases} Z_{LC} = \dfrac{s\frac{1}{C}}{s^2+\frac{1}{CL_1}} \\ Y_{AC} = \dfrac{1}{L_2 s} \end{cases}$$

$$T_{TOT} = \begin{bmatrix} - & - \\ \frac{s+\frac{14}{31}}{s}\cdot\frac{1}{R_g} & 1 \end{bmatrix} \cdot \begin{bmatrix} \frac{s^3+2s}{s(s^2+1/16)} & - \\ \frac{48}{31}\cdot\frac{1}{R_g}\frac{1}{s} & - \end{bmatrix}$$

- $R_g C = \left(\frac{s+\frac{14}{31}}{s}\right)\left(\frac{s^3+2s}{s(s^2+1/16)}\right) + (1)\left(\frac{48}{31}\cdot\frac{1}{s}\right)$

$$= \frac{s^3+2s^2+2s+1}{s(s^2+\frac{1}{16})}$$

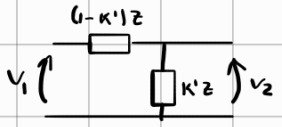
$$T(s) = \frac{1}{C} = \frac{R_g s(s^2+1/16)}{s^3+2s^2+2s+1} \Big/ K = R_g$$

b) $T(s) = \frac{K \cdot s(s^2 + 1/16)}{s^3 + 2s^2 + 2s + 1}$ / $K = K' R_g = -6\,dB$

- La transferencia es de la forma de un HP. Por lo que para obtener su ganancia en su banda de paso se debe evaluar en $s \to \infty$

  ↳ $\lim_{s \to \infty} T(s) = K' R_g = -6\,dB$

$\Longrightarrow$ $K' = \frac{1}{2R_g}$



$T(s) = \frac{K'\Omega}{K'\Omega + (1-K')\Omega} = K'$

